

Lecture 10: Regulatorstrukturer, IMC

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English version of slides by ©Bo Wahlberg



Today's Ordlista

English	Svenska
alternative control structures	alternativa regulatorstrukturer
cascade control	kaskadreglering
feedforward (also: feed forward)	framkoppling
inner/outer loop	inre/yttre loop
disturbance	störning



Outline

Recap

Cascade control

Feedforward

The Smith predictor

Internal Model Control

Other More Difficult Control Problems

In the previous lectures (F9, F11)..

- ▶ (F9) State feedback (from $x(t)$) and state feedback with observer (from $y(t)$)
- ▶ (F9) A systematic method to design a controller so that the closed-loop system has the “desired” poles (pole placement)
- ▶ (F11) Implementation (Euler, Tustin, Sampling Interval)

Today:

- ▶ Alternative control structures



Outline

Recap

Cascade control

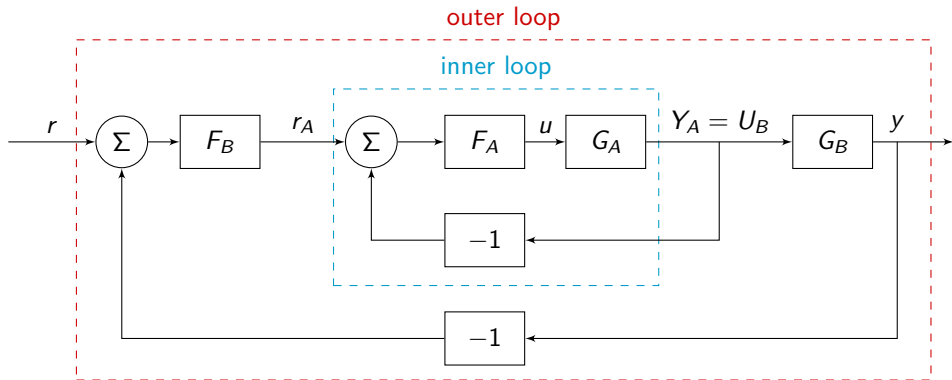
Feedforward

The Smith predictor

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Other More Difficult Control Problems

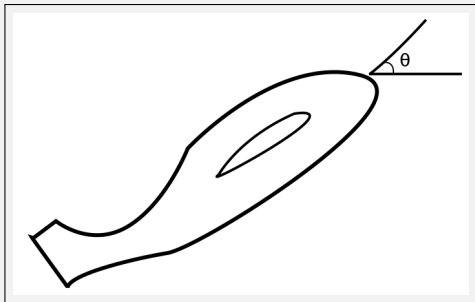
Cascade control



Similar to state feedback

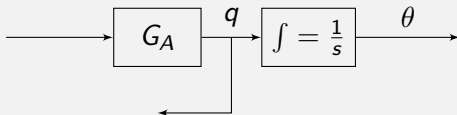
Cascade control

Example



$$\begin{cases} \theta & : \text{tilt angle (tippvinkel)} \\ \dot{\theta} = q & : \text{pitch rate (tippkastighet)} \end{cases}$$

Rate gyro feedback (rategyroåterkoppling):



Example (DC-motor)

$$G_A = \frac{1}{s+1}, \quad G_B = \frac{1}{s}$$

P-control: $F_A = K_A$, $F_B = K_B$

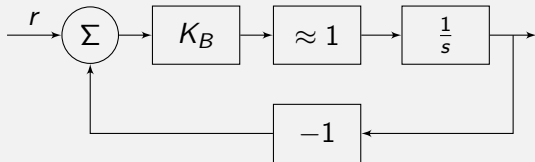
Inner loop: $Y_A = \frac{K_A}{s+1+K_A} R_A$ Choose a large K_A (e.g., more than 5 times faster than the outer loop) $\Rightarrow Y_A \approx 1 \cdot R_A$

Outer loop: $Y = \frac{K_B}{s+K_B} R$, choose $K_B < \frac{1+K_A}{5}$

Analysis:

$$Y = \frac{K_A K_B}{s^2 + (K_A + 1)s + K_A K_B} R$$

$L \sim (K_A K_B)$: the poles can be placed arbitrarily





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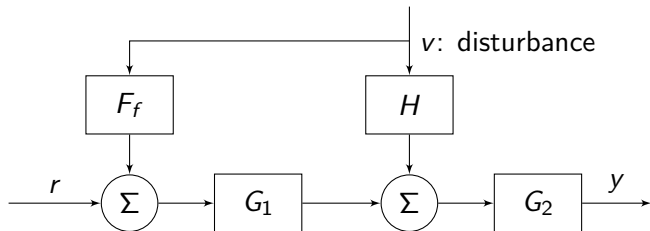
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Feedforward



$$Y = G_2(HV + G_1(R + F_f V)) = G_2(\underbrace{(H + G_1 F_f)}_{\text{set} = 0} V + G_1 R)$$

Eliminate the influence of v by choosing $H + G_1 F_f = 0$: $\Rightarrow F_f = -\frac{H}{G_1}$



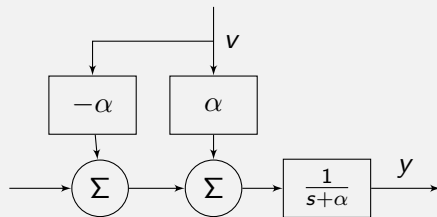
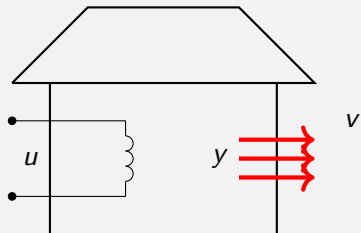
Assume that we have $\begin{cases} G_1 &= \frac{1}{s+1} \\ H &= 1 \end{cases}$.

One may, however, encounter problems with differentiating signals.

$$F_f(0) = -\frac{H(0)}{G_1(0)}$$

Combine with feedback!

Example (Climate/Temperature control from F1)



Does not wait until it gets cold indoors!



Outline

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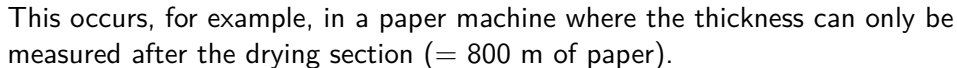
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We see that

Negative phase shift $\Rightarrow$ Deteriorates the phase margin (difficult to control).

A special-purpose controller

Step 1: Assume $T = 0$ and determine $\bar{F}$ such that G_c^d has good properties.

$$G_c^d(s) = \frac{\bar{F}(s)G_2(s)}{1 + \bar{F}(s)G_2(s)}$$

Step 2: Use the controller

$$F(s) = \frac{\bar{F}(s)}{1 + (1 - e^{-sT})\bar{F}(s)G_2(s)}$$

Step 3: The closed-loop G_c is identical to G_c^d except for the time delay. Note, however, that a known model (T and G_2) is used in the controller.

$$\Rightarrow G_c(s) = \frac{G(s)F(s)}{1 + G(s)F(s)} \stackrel{\{\text{calculations}\}}{=} \frac{\bar{F}(s)G_2(s)}{1 + \bar{F}(s)G_2(s)} e^{-sT} = G_c^d(s) e^{-sT}$$

Phase lead: Near ω_c

$$\Rightarrow \bar{F}(i\omega)G_2(i\omega) \approx -1 \quad \Rightarrow F(i\omega) \approx e^{+i\omega T} \bar{F}(i\omega)$$

Prediction of signals with frequencies near ω_c .



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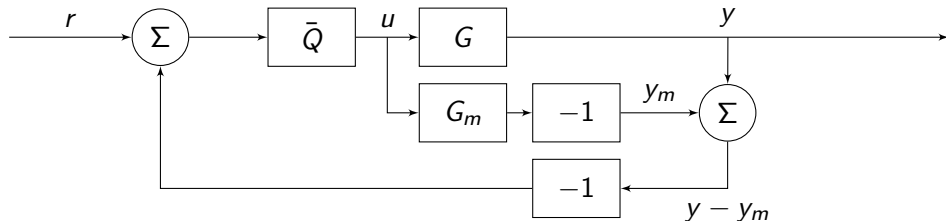
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Internal Model Control



$$U = \bar{Q}(\underbrace{R - Y}_E + G_m U) \Rightarrow U = \frac{\bar{Q}}{\underbrace{1 - \bar{Q}G_m}_{F, \text{ standard controller}}} E$$

$y_m = y$ if there are no model errors or disturbances ($G_m = G$)

$\Rightarrow$ Open-loop control $G_c = G\bar{Q}$!

Example (Otto Smith)

$$\bar{Q} = \frac{\bar{F}}{1 + \bar{F}G_2} \quad (\approx \frac{1}{G_2})$$

$$\Rightarrow G_c = \frac{\bar{F}G_2}{1 + \bar{F}G_2} e^{-sT} \quad (\approx e^{-sT})$$

$$\Rightarrow F = \frac{\bar{Q}}{1 - \bar{Q}G_m} \quad (= \text{"Otto Smith"})$$

In general,

$$\bar{Q} \approx \underbrace{1/\tilde{G}}_{\text{The part of } G \text{ that can be inverted}} \times G_c^d \Rightarrow G_c = G_c^d \frac{G}{\tilde{G}}$$

where G_c^d is the desired closed-loop system (which one cannot always achieve).
Time delays and unstable zeros cannot be eliminated!

Example (Otto-Smith)

$$G = G_2 e^{-sT} \Rightarrow \bar{Q} = \frac{\bar{F}}{1 + \bar{F} G_2} = \frac{G_c^d}{G_2}$$

Example

First order system

$$\text{model: } G_m = \frac{\bar{k}_m}{\bar{\tau}_m s + 1}, \quad \text{desired } G_c: G_c^d = \frac{1}{\lambda s + 1}$$

$$\bar{Q} = \frac{G_c^d}{G_m} \Rightarrow F = \frac{\bar{\tau}_m s + 1}{\bar{k}_m \lambda s}$$

Simple tuning of a PI controller with $K = \frac{\tau_m}{k_m \lambda}$, $T_I = \tau_m$

Second order system

$$\text{model: } G_m = \frac{\bar{k}_m}{(\bar{\tau}_1 s + 1)(\bar{\tau}_2 s + 1)}, \quad \text{desired } G_c: G_c^d = \frac{1}{\lambda s + 1}$$

Simple tuning of a PID controller, see course notes.

Disadvantages:

- ▶ Cannot handle unstable systems, “open-loop control”
- ▶ Sensitive to disturbances

Perfect control $G_c = 1 \cdot e^{-sT}$.

Sensitivity function $S(i\omega) = 1 - e^{-i\omega T}$.

Note that $S(i\omega) = 1 - (-1) = 2$ when $\omega = \frac{\pi}{T}$

Amplifies disturbances for $\omega \approx \frac{\pi}{T}$.

Fundamentally difficult problem!

Reduce ω_c relative to T !

EL2520 Reglerteknik, fortsättningskurs i Period 4!



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More difficult control problems:

- Zeros in the high-frequency part (non-minimum phase system)

Example

$$\underbrace{\frac{1}{s+1}}_{\text{slow}} - \underbrace{\frac{2}{s+10}}_{\text{fast}} = \frac{(-s+8)}{(s+1)(s+10)} = \frac{(s+8)}{(s+1)(s+10)} \left[\underbrace{\frac{(-s+8)}{(s+8)}}_{\text{negative phase shift}} \right]$$

Compare with

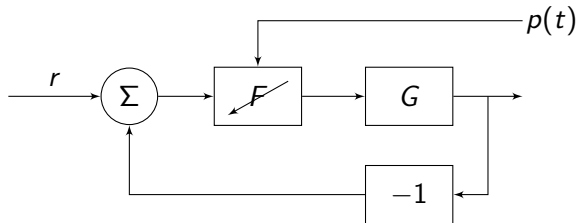
$$e^{-\frac{s}{4}} = \frac{e^{-s/8}}{e^{s/8}} \approx \frac{1 - \frac{s}{8}}{1 + \frac{s}{8}}$$

Other Control Problems

► Multivariable systems

- Multiple input and output signals $\Rightarrow$ Control, feedback, and optimization (optimal control)

► Time-varying systems with gain-scheduling

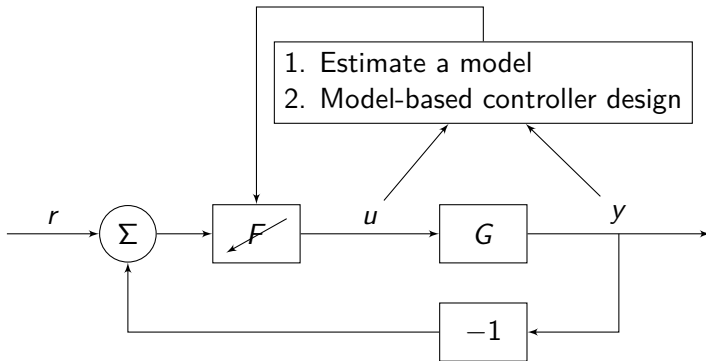


F changes using external variables $p(t)$

For example, for flight paths.

Other More Difficult Control Problems

► Adaptive control



F changes using the estimated model

- Autotuning (“adaptive control during startup”)
- Reinforcement learning

AlphaGo, Pluribus (poker), **tic-tac-toe**, ...



- ▶ Possible to analyze all possible games
- ▶ Perfect information
- ▶ Does Reinforcement Learning work here (EL2805)?
- ▶ Learn to win by playing (without thinking)
- ▶ Always ends in a tie if both play optimally

Trade-off between exploration and exploitation

- ▶ **A policy** is a mapping from perceived states of the environment to actions to be taken when in those states
- ▶ **A reward signal** defines the goal in a reinforcement learning problem. On each time step, the environment sends to the reinforcement learning agent a single number, a reward. The reward signal indicates what is good in an immediate sense (win, lose, or tie in one game)
- ▶ **The value function** specifies what is good in the long run (over many games)

Reinforcement Learning: An Introduction by Richard S. Sutton and Andrew G. Barto

Course EL2805 Reinforcement Learning

Do as usual or try new ideas (sometimes)

- ▶ Reward: +100 for a passed exam
- ▶ -1000 for a missed exam
- ▶ **Policy:** For example, learn each page of the book by heart, solve all exercises, memorize all past exams,..
- ▶ That is too much work, there must be an easier way!
- ▶ Read the course information, you are allowed to use the book during the exam, attend exercises, study sessions, etc.


$$\dot{y}(t) = -ay(t) + bu(t)$$

Minimize the cost function (finite horizon) with respect to the input

$$J = \int_0^t [y^2(\tau) + q_1 u^2(\tau)] d\tau + q_2 y^2(t)$$

$$J = \int_0^\infty [y^2(\tau) + qu^2(\tau)] d\tau$$
$$u(t) = -K(t)y(t)$$

$$u(t) = -Ky(t), \quad \text{infinite horizon}$$


$$J = \int_t^{t+\Delta} [y^2(\tau) + q_1 u^2(\tau)] d\tau + q_2 y^2(t + \Delta)$$
$$|y(t)| \leq \epsilon_y \quad |u(t)| \leq \epsilon_u, \quad \forall t$$

MPC: Use only the optimal solution $u(t)$ for $t \leq t + \delta$ and then compute a new solution at the next time step $t \mapsto t + \delta$

Easier for sampled systems (Lecture 11)

Course EL2700 Model Predictive Control